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DSCI 552 HW 3

ISLR 3.7.4

a. Since cubic regression models have more variance, they are more able to adjust to even the smallest deviations away from linear in the training data than linear regression models. In the event that the data is perfectly linear, then cubic and linear regression training RSS will be identical; otherwise, the cubic regression model will have a smaller training RSS. Therefore, we expect that the cubic training RSS will be lower.
b. Since the true relationship between X and Y is linear, a linear regression model will perform better on unseen (i.e., test) data than a cubic regression model. Therefore, we expect that the linear test RSS will be lower.
c. Similarly to part (a), the cubic model will always be more able to track the data points than a linear model, particularly if we know that the true relationship between X and Y is not linear. Therefore the cubic training RSS will again be lower.
d. Since we don't know exactly how the data deviates from a linear relationship, we do not have enough information to predict whether the linear or cubic test RSS will be lower. If the data is closer to linear than cubic, then the linear test RSS will be lower; if it is closer to cubic than linear, then the cubic test RSS will be lower. Therefore we do not know given the information available to us.